

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 8: Remainder and Factor Theorems

Previous year board questions, topic-wise: 9 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 The factor theorem

1. Which factor is common to $x^2 - 4$ and $x^3 - x^2 - 4x + 4$? (a) $(x + 1)$ (b) $(x - 1)$ (c) $(x - 2)$ (d) $(x - 4)$ **ICSE 2025, 1 mark**

2 Factorising a cubic polynomial

2. $(x - 2)$ is a factor of $2x^3 - 7x^2 + kx - 2$. Find k , and then factorise the polynomial completely. **ICSE 2025, 3 marks**
3. $(x - 1)$ is a factor of $3x^3 + 8x^2 - 15x + k$. Find k , and then factorise the polynomial completely. **ICSE 2024, 3 marks**
4. Use the factor theorem to factorise $2x^3 - x^2 - 13x - 6$ completely. **ICSE 2023, 4 marks**
5. Use the factor theorem to factorise $6x^3 + 17x^2 + 4x - 12$ completely. **ICSE 2020, 3 marks**

3 The remainder theorem

6. What must be subtracted from $x^3 + x^2 - 2x + 1$ so that the result is exactly divisible by $(x - 3)$? (a) -31 (b) -30 (c) 30 (d) 31 **ICSE 2024, 1 mark**
7. What must be added to $2x^3 - 3x^2 - 8x$ so that it leaves a remainder of 10 when divided by $2x + 1$? **ICSE 2020, 3 marks**

4 Finding unknown constants

8. $(x - 2)$ is a factor of $x^3 - kx - 12$. Find k . (a) 3 (b) 2 (c) -2 (d) -3 **ICSE 2023, 1 mark**
9. Find a if $(x - a)$ is a factor of $3x^3 + x^2 - ax - 81$. **ICSE 2023, 4 marks**

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